R1/XCON

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Agenda

01 History

02 Problem Addressed

03 What is the expertise that it mimics?

04 What are the advantages of the expert system?

05 Challenges

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History

John P. McDermott of CMU created the R1 program, also known internally as XCON, in OPS5 in 1978 to help with the ordering of DEC's VAX computer systems by automatically choosing the computer system components based on the customer's requirements.

McDermott employed specialists from DEC's PDP/11 and VAX computer systems units in the creation of the system. These specialists occasionally clashed about the ideal setup amongst themselves. The ensuing "sorting it out" provided an added advantage in terms of the caliber of the delivered VAX systems.

The Salem, New Hampshire, facility of DEC was the first to employ XCON. At the end, it contained roughly 2500 regulations. With 80,000 orders handled by 1986, it had a 95-98% accuracy rate. By minimizing

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The Salem, New Hampshire, facility of DEC was the first to employ XCON. At the end, it contained roughly 2500 regulations. With 80,000 orders handled by 1986, it had a 95–98% accuracy rate. By minimizing the need for consumers to receive free components when technicians made mistakes, by hastening the assembly process, and by raising customer satisfaction, it was anticipated that DEC would save $25M annually.

Agenda

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Problem Addressed

01 Customers had to order individual parts when purchasing a VAX, which was time-consuming and confusing.

02 Salespeople at DEC were not always knowledgeable, resulting in incomplete orders.

03 Customers received incomplete orders, causing delays, frustration, and even legal action.

04 XCON was created to automate and simplify the ordering process.

05 XCON interacted with salespeople to ensure complete and accurate orders.

06 XCON provided a clear and concise system specification/order slip for customers, preventing delays and customer dissatisfaction.

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What is the expertise that it mimics?

XCON is used to validate the technical correctness (configurability) of customer orders and to guide the actual assembly of these orders. It provides the following functionality:

- Configures CPUS, memory, boxes, backplanes, cabinets, and power supplies, disks, tapes., HSC/CI, printers, etc.
- Diagrams complete system configuration (Figure 1 overlays selected pages of a sample XCON diagram)
- Checks marketing restrictions, system building block menus, and prerequisites
- Assigns addresses/vectors and determines box power status
- Partitions multiple-cpu orders and cluster systems
- Determines and lists cabling information
- Lists components ordered with configuration-related comments
- Generates warning messages on issues affecting technical validity.

Agenda

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05 Challenges

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02 Problem Addressed

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What are the advantages of the expert system?

The use of the configuration systems ensures that complete, consistently configured systems are shipped to the customer. Incomplete orders do not get through the process. In addition, XCON generates configurations which optimize system performance, so customers consistently get the best view of our products.

- Configures CPUs, memory, cabinets, power supplies, disk, tapes and so on.
- Determines and list cabling information.
- List components ordered with configuration related comments.
- Generate warning messages on issues affecting technical validity.

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Challenges

The main challenges stemming from the emerging nature of this technology are:

01

Dealing creatively with the current limitations of the specialized languages and tools

02

Formulating an appropriate software engineering methodology (this is the thrust of the RIME development effort) to produce efficient, maintainable, extensible expert systems.

Today's expert systems languages (and shells) offer varied but somewhat limited functionality. Most do not provide true extensibility to a production environment. There are very few knowledge engineering productivity tools available as yet on the market.

Challenges

The main challenges stemming from the emerging nature of this technology are:

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Challenges

Bachant and Soloway introduce a software engineering methodology appropriate for the development of expert systems. They make two primary points:

01 A task such as XCON's, where thousands of highly context-dependent decisions must be made in order to correctly configure a system, can take advantage of a language, like OPS5, that presupposes "situation recognition" rather than "algorithmic" control.

02 But, as the number of alternatives that an expert system must decide among becomes large, there is strong motivation to augment situation recognition with other kinds of control mechanisms

Thank You

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